

United International University

QUEST FOR EXCELLENCE

ASSIGNMENT

Computer Network Laboratory

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Part 1: Introduction

1.1 Overview of Socket Programming

A socket is a communication endpoint that enables data exchange between two processes over a network. Socket programming is widely used to develop network-based applications such as web servers, email systems, chat applications and file-sharing services.

In the client-server architecture, the server waits for incoming requests while the client initiates communication. Each socket is identified by an IP address and a port number allowing devices to communicate efficiently.

1.2 Objective of the Project

The objective of this project is to develop a basic HTTP server using Python socket programming. The server listens on port 8080 and serves HTML pages requested through a web browser.

The server performs the following tasks:

1. Creates a TCP socket.
2. Binds the socket to localhost and port 8080.
3. Waits for browser requests.
4. Processes the requested URL path.
5. Returns the corresponding HTML file.
6. Displays a 404 error page for invalid requests.

Part 2: Implementation

2.1 Server Program

The server is developed using Python's built-in socket module. It continuously listens for incoming browser requests and sends appropriate HTTP responses.

Server Code :

```
import socket
```

```

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

server.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)

server.bind(('127.0.0.1', 8080))

server.listen(5)

print("Server running...")
print("Visit http://127.0.0.1:8080")

while True:

    client, addr = server.accept()

    request = client.recv(1024).decode()

    print("Request received:")
    print(request)

    try:
        path = request.split()[1]

        if path == '/':
            filename = "index.html"
        else:
            filename = path[1:]

        with open(filename, "r") as file:
            content = file.read()

        response = "HTTP/1.0 200 OK\r\n\r\n"
        response += content

    except FileNotFoundError:
        response = """
HTTP/1.0 404 NOT FOUND\r\n\r\n
<html>
<body>
<h1>404 Not Found</h1>
</body>
</html>
"""

    client.send(response.encode())
    client.close()

```

a web browser is opened and the URL `http://localhost:8080/` is accessed. The server receives the browser request and returns the contents of the `index.html` file . Similarly, when the URL `http://localhost:8080/about.html` is requested, the server loads and displays the contents of `about.html`. The same process applies to `contact.html` and any other valid HTML file present in the project directory.

If a requested file does not exist, such as:

`http://localhost:8080/abc.html`

the server sends the following response:

HTTP/1.0 404 NOT FOUND

and displays a custom "404 Not Found" page in the browser.

This demonstrates that the HTTP server can successfully handle browser requests, serve multiple HTML pages, and return appropriate responses for both valid and invalid URL paths.

2.2 Index page:

```
<!DOCTYPE html>
<html>
<head>
  <title>Home Page</title>
</head>
<body>
  <h1>Welcome to Index Page</h1>
  <p>This page is loaded from index.html</p>
</body>
</html>
```

2.3 About page

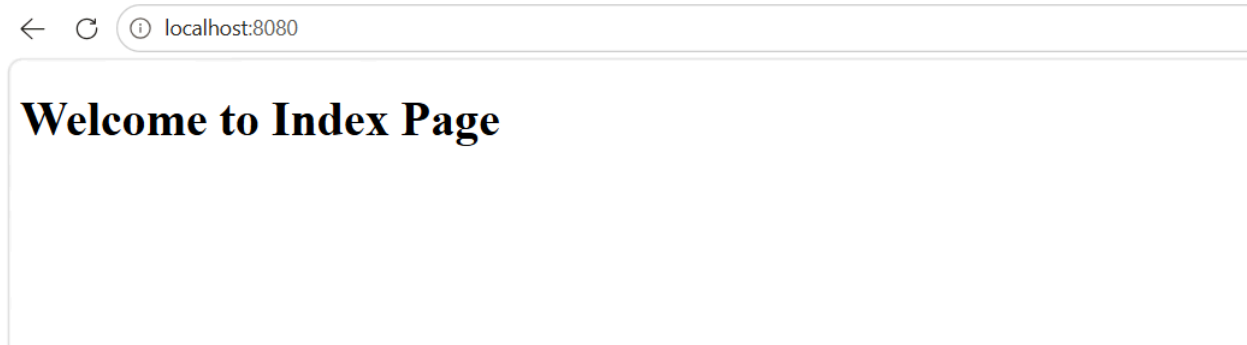
```
<!DOCTYPE html>
<html>
<head>
  <title>About Page</title>
</head>
<body>
  <h1>About Page</h1>
  <p>This page is loaded from about.html</p>
</body></html>
```

Part 3:(Screenshots)

3.1 Starting Server:

```
Command Prompt - .venv\Scr x + v
C:\Users\HP>cd desktop
C:\Users\HP\Desktop>cd setu server
C:\Users\HP\Desktop\setu server>.venv\Scripts\python.exe server.py
Server running...
Visit http://localhost:8080
```

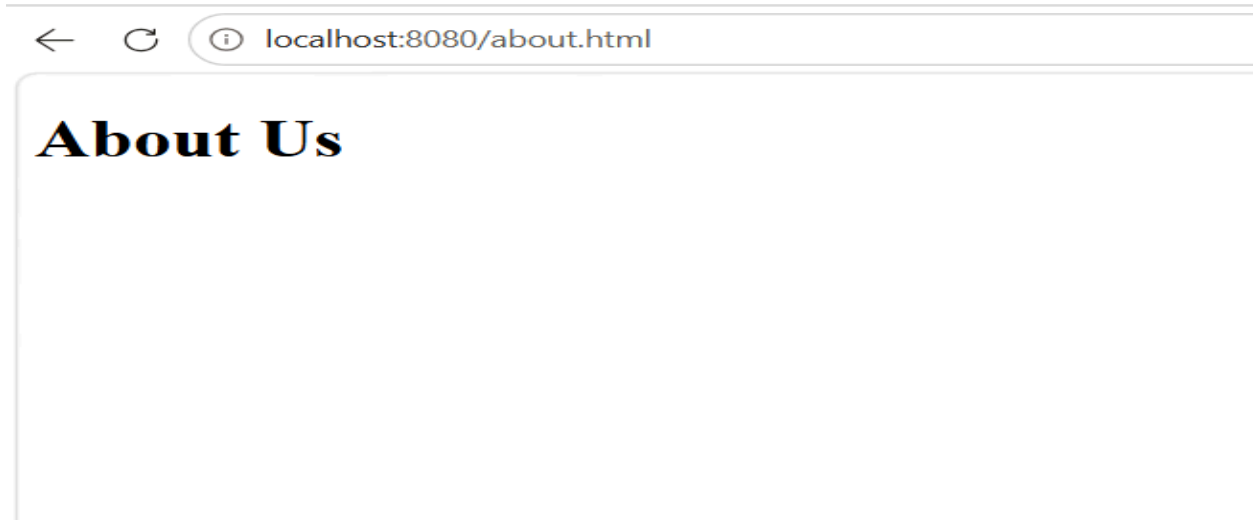
3.2 Accessing base file(http://localhost:8080/):



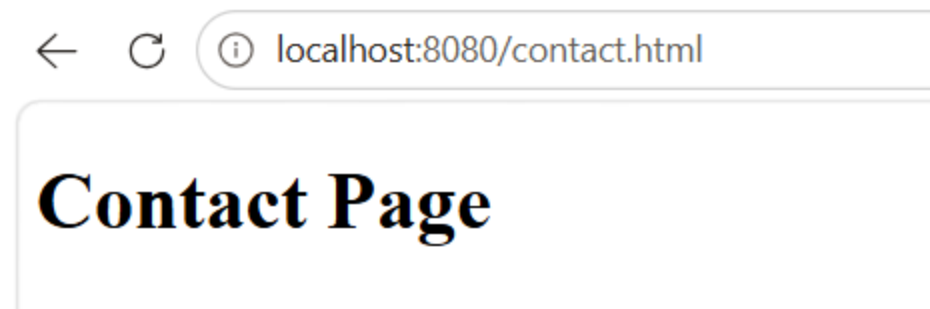
3.3 After acceptance, server:

```
Command Prompt - .venv\Scr x + v
C:\Users\HP>cd desktop
C:\Users\HP\Desktop>cd setu server
C:\Users\HP\Desktop\setu server>.venv\Scripts\python.exe server.py
Server running...
Visit http://localhost:8080
Request received:
GET / HTTP/1.1
Host: 127.0.0.1:8080
Connection: keep-alive
sec-ch-ua: "Google Chrome";v="149", "Chromium";v="149", "Not)A;Brand";v="24"
sec-ch-ua-mobile: ?0
sec-ch-ua-platform: "Windows"
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/149.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Sec-Fetch-Site: none
Sec-Fetch-Mode: navigate
Sec-Fetch-User: ?1
Sec-Fetch-Dest: document
Accept-Encoding: gzip, deflate, br, zstd
Accept-Language: en-US,en;q=0.9
```

3.4 Accessing <http://localhost:8080/about.html> :



3.5 Accessing <http://localhost:8080/contact.html> :



3.6 Searching a file that doesn't exist localhost:8080/abc.html :

